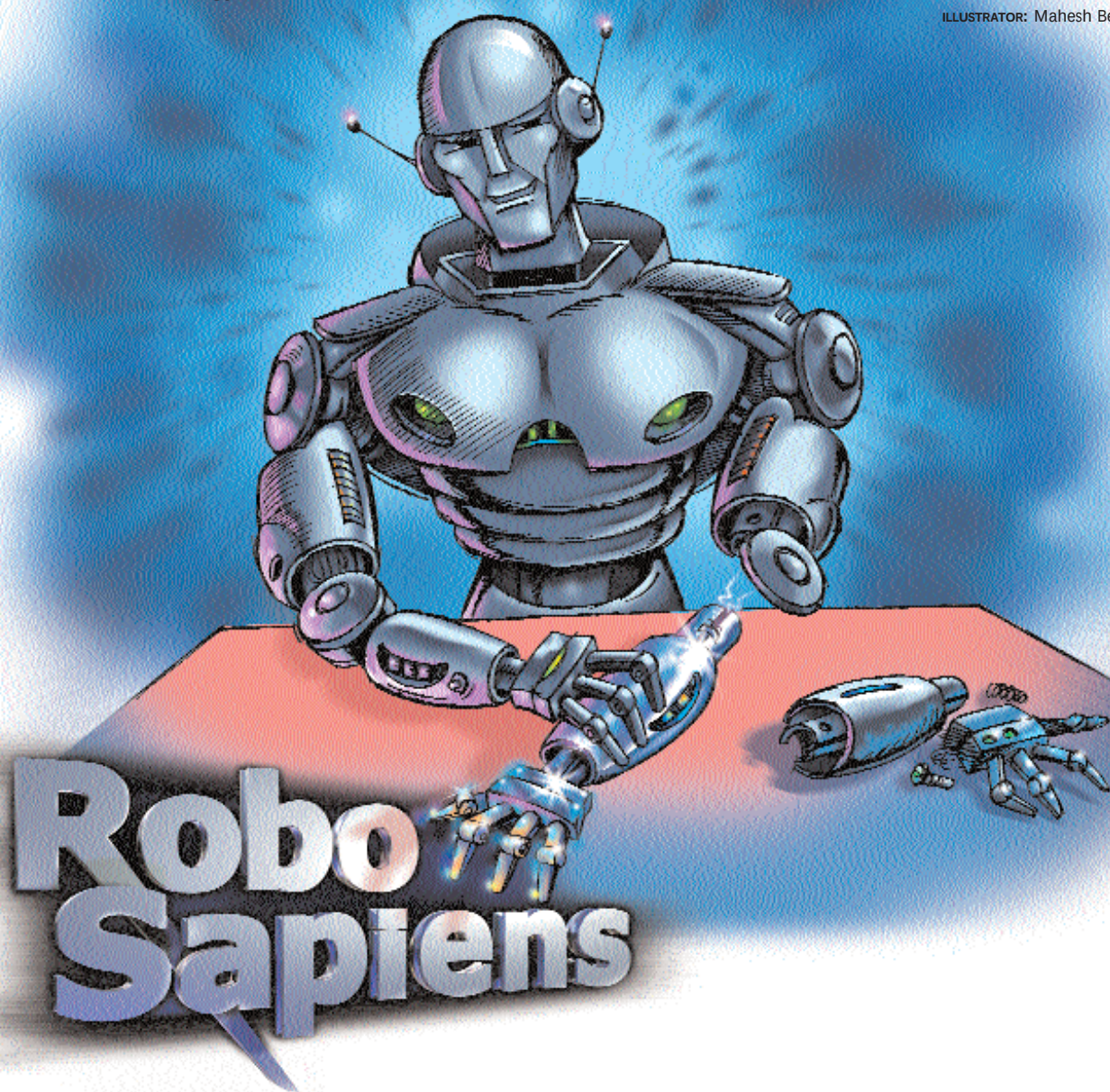




The new wave of reconfigurable robots will learn, change themselves, and evolve

Do a Google search on 'robots will', and you see: 'Robots will steal your job...', 'Robots will kill...', 'Robots will change us...', 'Robots will make good friends...', 'Robots will manage your network...', etc. Clearly, people have a lot of beliefs about what robots will do, and this is justified because robots, or bots, already do so many amazing things. They give back massages, vacuum entire rooms and play soccer with each other. Not many people are aware that we are on the cusp of a new wave of artificial intelligence, which promises robots that will learn about their environment, change as the environment demands, improve themselves and, possibly, reproduce. Whatever else robots may or may not

do, we're sure of a couple of things—they will learn; they will change themselves; they will evolve by themselves.

Roach bots

Robotics is rapidly embracing ideas from other disciplines, leading to new techniques and sub-technologies. Honda has its Asimo humanoid, and Sony has its robotic Aibo dog, both of which are impressive. However, the Asimo and the Aibo are 'static' bots in the sense that they do not dynamically adapt to their environment; what they learn is only facts that are added to their knowledge base. These facts do help them function better, but cannot help them evolve in any way. Asimo and Aibo-style research will not drive the future; it's evolutionary

and animal-mimicking techniques that will dominate. Animal-inspired robotics aims at emulating certain animals, especially insects and the like. Biologists and engineers have worked together at producing bot versions of popular insects such as flies and cockroaches. If you're afraid of cockroaches, then the three bots at Case Western Reserve University, Ohio, United States, are your worst nightmare: they're huge, they look like cockroaches and they do the things that cockroaches do—jump, fly, and so on. The goal of this experiment was to build machines that could go where humans won't—inside nuclear reactors, or into a fire. The inspiration from biology is driven by laziness: why engineer a creature when nature has already done it for us?